

MID TERM EXAMINATION

Course: Data Structures (CS101)

Set: A | Time: 3 Hours | Max Marks: 40 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
ALL

SECTION A: MCQs

1. Which of the following is NOT typically considered a primary characteristic or benefit of a well-designed Database Management System (DBMS)? **[2 Marks]**

- a) Reduced data redundancy
- b) Enhanced data security
- c) Increased data inconsistency
- d) Improved data sharing

2. In a relational database, a column or a set of columns that uniquely identifies each row in a table is known as **[2 Marks]**

- a) Foreign key
- b) Candidate key
- c) Primary key
- d) Composite key

3. Which data model represents data in a tree-like structure, where data is organized into parent-child relationships, and a child can only have one parent? **[2 Marks]**

- a) Relational model
- b) Network model
- c) Hierarchical model
- d) Object-oriented model

4. The process of organizing the columns and tables of a relational database to minimize data redundancy and improve data integrity is called: **[2 Marks]**

- a) Indexing
- b) Normalization
- c) Query optimization

d) Concurrency control

5. Which component of a DBMS is responsible for receiving user requests (e.g., SQL queries) and converting them into an efficient execution plan for the database engine? **[2 Marks]**

- a) Storage Manager
- b) Transaction Manager
- c) Query Processor
- d) Buffer Manager

6. When multiple transactions are executing concurrently, which ACID property ensures that the outcome is equivalent to some serial execution of those transactions, preventing interferences? **[2 Marks]**

- a) Atomicity
- b) Consistency
- c) Isolation
- d) Durability

SECTION B: SHORT ANSWERS

7. Briefly explain the primary purpose of a Database Management System (DBMS). **[4 Marks]**

8. Differentiate between data redundancy and data inconsistency in a database system. **[5 Marks]**

9. List and briefly describe two significant advantages of using a DBMS over a traditional file processing system. **[6 Marks]**

10. Explain the concept of data independence in a DBMS. Differentiate between logical and physical data independence. **[7 Marks]**

11. What are the ACID properties in the context of database transactions? Briefly describe two of these properties. **[6 Marks]**

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